

BASE CARLINI, Antarctica

WMO: 890530

Lat: 62.238S

Lon: 58.665W

Elev: 36

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.1	2.8	-7.7	3.6	-1.6	-2.9	4.7	3.4	63.5	31.6	56.0	24.7	22.0	50	1.277

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.9	43.1	40.0	41.4	38.8	40.4	38.0	40.5	42.3	39.4	40.9	38.5	39.9	22.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
38.7	34.6	40.8	37.8	33.4	39.8	36.9	32.2	38.9	15.6	42.6	15.1	41.2	14.7	40.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
53.5	47.9	44.2	-4.9	49.3	6.0	4.1	-9.2	52.3	-12.7	54.7	-16.0	57.0	-20.4	60.0		
			-5.6	43.9	5.6	2.0	-9.6	45.4	-12.8	46.6	-16.0	47.7	-20.0	49.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.1	36.0	35.8	33.9	29.4	27.8	23.2	21.5	21.9	24.8	28.9	31.9	34.2
	DBStd	7.92	2.55	3.04	4.26	6.24	6.23	8.11	8.99	8.67	7.64	4.99	3.68	2.69
	HDD50	7639	434	399	498	618	690	805	884	870	757	653	542	489
	HDD65	13114	899	819	963	1068	1155	1255	1349	1335	1207	1118	992	954
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	21.4	18.0	19.6	21.2	21.3	21.4	21.9	21.9	22.7	24.0	24.4	21.7	18.7
Precipitation	PrecAvg	40.9	3.2	3.2	4.7	4.0	3.8	3.0	2.8	3.0	3.2	3.9	3.3	2.8
	PrecMax	54.8	4.5	4.9	7.4	6.0	6.3	4.5	4.3	5.6	5.4	6.7	4.5	4.5
	PrecMin	33.7	1.9	1.9	2.8	2.3	1.7	1.6	1.3	0.6	1.1	2.0	1.2	1.2
	PrecStd	4.6	0.6	0.7	1.1	0.9	1.1	0.7	0.9	1.1	0.9	1.0	0.8	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	44.5	44.3	41.8	40.2	38.8	37.0	36.8	37.9	40.4	43.4	43.8
		MCWB	41.3	41.5	40.8	39.2	37.4	36.3	34.7	34.8	35.8	37.3	39.4	40.3
	2%	DB	42.5	42.1	41.6	39.4	38.5	36.6	35.3	35.0	36.1	37.7	39.2	41.1
		MCWB	39.6	39.5	39.2	37.5	36.3	34.5	33.6	33.4	34.6	35.6	36.8	38.3
	5%	DB	41.1	40.8	40.2	37.9	37.1	35.0	34.1	33.8	35.0	36.3	37.8	39.3
		MCWB	38.5	38.4	38.2	36.2	35.4	33.3	32.7	32.5	33.8	34.5	35.8	36.8
	10%	DB	39.8	39.6	39.1	36.8	35.7	33.5	32.7	32.6	34.0	35.1	36.6	38.0
		MCWB	37.5	37.5	37.4	35.3	34.3	32.2	31.5	31.6	32.9	33.6	34.9	35.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	41.9	41.4	39.5	38.3	36.7	35.4	35.5	36.7	37.9	40.0	41.1
		MCDB	44.5	43.8	43.2	40.9	39.3	38.3	36.2	36.2	37.3	39.3	43.1	42.9
	2%	WB	40.1	40.0	39.6	37.8	36.7	34.8	34.0	33.8	35.2	35.9	37.5	38.6
		MCDB	41.9	41.6	41.0	38.9	37.9	36.2	34.9	34.6	35.8	37.1	38.7	40.5
	5%	WB	38.8	38.9	38.6	36.6	35.7	33.5	32.9	32.7	34.0	34.8	36.2	37.2
		MCDB	40.4	40.3	39.8	37.7	36.9	34.6	33.9	33.5	34.8	36.0	37.3	38.9
	10%	WB	37.8	38.0	37.7	35.5	34.5	32.2	31.6	31.6	33.0	33.8	35.2	36.1
		MCDB	39.4	39.2	38.9	36.6	35.6	33.4	32.7	32.5	33.9	34.9	36.3	37.5
Mean Daily Temperature Range	5% DB	MDBR	4.9	4.9	5.5	6.3	7.1	8.4	9.3	8.5	7.9	5.9	5.5	4.9
		MCDBR	7.1	6.1	6.2	6.2	7.4	8.1	8.2	7.8	6.9	6.7	7.5	6.8
		MCWBR	5.3	4.8	5.2	5.7	6.5	7.3	7.8	7.4	6.5	5.6	6.1	5.2
	5% WB	MCDBR	6.6	5.4	5.8	5.9	6.9	8.0	7.7	7.8	6.7	6.4	7.0	6.4
		MCWBR	5.2	4.7	5.3	5.7	6.3	7.6	7.8	7.5	6.6	5.7	6.0	5.2
Clear-Sky Solar Irradiance	taub	0.308	0.308	0.297	0.279	0.223	0.366	0.209	0.267	0.287	0.309	0.304	0.310	
	taud	2.494	2.501	2.502	2.374	2.097	2.834	2.044	2.237	2.369	2.400	2.456	2.463	
	Ebn at Noon	307	291	267	215	141	96	139	210	263	286	306	310	
	Edh at Noon	31	28	23	16	10	6	10	18	25	30	32	33	
All-Sky Solar Radiation	RadAvg	1601	1155	620	271	78	33	56	201	531	986	1512	1813	
	RadStd	145	87	52	21	7	1	3	17	42	65	91	145	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.06	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	-0.47	-0.76	N/A	N/A	N/A	N/A

Nomenclature: See separate page